

Faculty of Computing

BSc (Hons) in Information Technology

Year 4 Semester 1 & 2

SE4041 - Mobile Application Design and Development (MADD) **Assignment 01 – Kotlin**

Objective

Students are required to develop a mobile application using Kotlin by following the Software Development Life Cycle (SDLC) process. The stages of Deployment and Maintenance are excluded from this assignment.

The assignment aims to assess students' ability to plan, design, develop, and test a mobile application effectively.

SDLC Stages to be Followed

- Planning & Requirement Analysis
- Defining Requirements
- Design
- Development
- Testing

Topic Allocation

Each student will be assigned a specific topic based on the last digit of their registration number.

Last Digit of Registration Number	Topic
0	AI-Powered Personal Productivity Mobile Application
2	Smart Mental Wellness and Habit Tracking Mobile Application
4	Personal Finance and Expense Tracking Mobile Application
6	Smart Sustainability and Carbon Footprint Tracking Mobile Application
8	Community Services and Local Marketplace Mobile Application

Detailed Requirements

Phase 01 – Planning & Requirement Analysis

Propose Your Idea to the Panel and Obtain Approval

Submission: Report Submission to Course Web

Evaluation:

Students should:

- Identify the problem addressed by the application.
- Identify the target audience.
- Conduct a feasibility study.
- Define the scope of the application.

Defining Requirements

Students should:

- Identify functional requirements.
- Identify non-functional requirements.
- Define the features and functionalities of the application.

Design UI Prototype using Figma (10 Marks)

Submission: Report Submission to Course Web

Evaluation:

Students should design a complete UI prototype using Figma.

Requirements:

- Ideation
- Application of the 60-30-10 Color Rule
- Main Interface Design
- Design Aesthetics and Usability

Development using Kotlin (15 Marks)

Submission: Complete Application via GitHub

Evaluation:

Students should develop mobile applications using Kotlin and Android Studio.

Requirements:

- Appropriate Layout Selection
- Proper UI Components
- User Interactivity
- Core Functionalities
- Database Integration
- GitHub Repository Management

Students must maintain their GitHub repository from the beginning of the project with regular commits throughout the development process.

Testing

Students should:

- Perform Unit Testing.
- Perform Integration Testing.
- Document the testing process and results.

Documentation (5 Marks)

Submission: Report Submission to Course Web

Evaluation:

The report should include:

- Overview of the Application
- Features and Functionalities
- Challenges Faced
- Application Screenshots
- Testing Procedures and Results

Students must make their GitHub repository public and include the repository link in the report.

Marking Scheme

Design UI Prototype using Figma (10 Marks)

Criteria	Excellent	Very Good	Good	Average	Needs Improvement
Ideation (Max 3 Marks)	Highly innovative idea that clearly addresses the assigned domain and user needs	Creative and relevant ideas with strong justification	Appropriate idea with adequate justification	Basic idea with limited originality	Unclear or poorly defined idea
Applying 60-30-10 Rule (Max 2 Marks)	Rules applied consistently and effectively throughout the design	Minor inconsistencies in color distribution	Mostly follows the rule	Limited application of the rule	Rule not applied correctly
Main Interface Design (Max 3 Marks)	Professional, intuitive, and well-structured interface	Well-designed interface with minor improvements needed	Functional and organized design	Basic design with usability concerns	Poorly organized interface
Design Aesthetics and Usability (Max 2 Marks)	Excellent visual appeal and user experience	Good visual consistency and usability	Acceptable aesthetics and usability	Limited consideration for usability	Poor aesthetics and difficult navigation

Development using Kotlin (15 Marks)

Criteria	Excellent	Very Good	Good	Average	Needs Improvement
Selecting Suitable Layouts (Max 2 Marks)	Layouts are appropriate, responsive, and well-structured	Mostly appropriate layouts used	Adequate layout selection	Basic layout implementation	Poor layout selection
Using Proper Views for the User Interface (Max 2 Marks)	UI components are selected and used effectively throughout	Good use of appropriate views	Adequate use of views	Limited use of suitable views	Incorrect or poor use of views
Interactivity (Max 1 Mark)	Application interactions are smooth and responsive	Good interactivity with minor issues	Functional interactions	Limited interactivity	Little or no interactivity
Functionality (Max 4 Marks)	All required features are fully implemented and working correctly	Most features implemented correctly	Core functionality implemented	Partial functionality available	Major functionality missing
DB Connection (Max 3 Marks)	Database is fully integrated and functions correctly	Good database implementation	Adequate database functionality	Limited database support	Database missing or not functional
GitHub Usage (Max 3 Marks)	Regular commits with excellent repository management	Good commit history maintained	Adequate use of version control	Few commits recorded	Poor repository management

Documentation (5 Marks)

Criteria	Excellent	Very Good	Good	Average	Needs Improvement
Overview (Max 1 Mark)	Comprehensive and clear overview of the application	Well-written overview	Adequate overview	Basic overview	Incomplete overview
Features (Max 1 Mark)	Features are clearly described and justified	Features well explained	Adequate feature descriptions	Limited descriptions	Features not properly described
Challenges (Max 1 Mark)	Challenges and solutions are critically discussed	Good discussion of challenges	Adequate discussion	Limited discussion	Challenges not discussed
Screenshots (Max 1 Mark)	Clear, relevant, and properly labelled screenshots	Good screenshots provided	Adequate screenshots	Limited screenshots	Missing screenshots
Testing (Max 1 Mark)	Comprehensive testing evidence and results documented	Good testing documentation	Adequate testing evidence	Limited testing evidence	Testing not documented